Calculating Determinants.

Cramer's rule not in final.

→ 3 methods:

① Pivots

$$A = LU$$

$$|A| = |L|\,|U|$$

$$\left(1 \times p_1 p_2 p_3 \cdots p_n\right)$$

② The Big Formula

$$\begin{vmatrix} a & c \\ b & d \end{vmatrix} = ad - bc$$

$$\begin{vmatrix} a+0 & c+0 \\ \end{vmatrix}$$

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

1st row change:

$$= \begin{vmatrix} a & 0 \\ b+0 & d+0 \end{vmatrix} + \begin{vmatrix} 0 & c \\ b+0 & d+0 \end{vmatrix}$$

$$= \begin{vmatrix} a & 0 \\ b & 0 \end{vmatrix} + \begin{vmatrix} a & 0 \\ 0 & d \end{vmatrix}$$

$$+ \begin{vmatrix} 0 & c \\ b & 0 \end{vmatrix} + \begin{vmatrix} 0 & c \\ 0 & d \end{vmatrix}$$

diagonal Mat (upper so $P_1P_2 \leftarrow$ ad is det)

to make upper, permutation needed which gives $-\det = -bc$

$$= ad - bc$$

$n \times n$ $\quad n \to 2$

all relevant terms have only 1 non zero element, others are easy.

$3 \rightarrow ?$

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \rightarrow 2 \text{ choices}$$

$$\begin{bmatrix} a_{11} & 0 & 0 \\ 0 & 0 & a_{23} \\ 0 & a_{32} & 0 \end{bmatrix} = a_{11}a_{22}a_{32} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

1 swap to get I

$$= -a_{11}a_{22}a_{32}$$

$3 \times 2 \times 1 = 6$ choices.

only these 2 possible by choosing a_{11} 1st.

$$\det(A) = a_{11}a_{22}a_{33} - a_{11}a_{23}a_{32} + \cdots$$

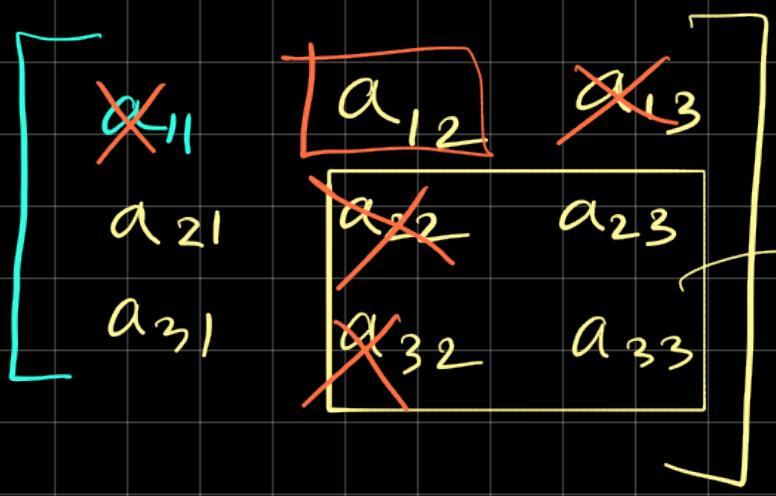
$$= a_{11}(a_{22}a_{33} - a_{23}a_{32}) + a_{12}(\quad)$$

$\det$(of remaining matrix)

$\det(12)$

$+ a_{13}(\quad)$

det c_{13}



$C_{12} \rightarrow \begin{bmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{bmatrix}$

C_{11} $a_{21}a_{33} - a_{31}a_{23}$

③ Co factors

$$\det(C_{ij}) = (-1)^{i+j}\left(\begin{array}{l}\text{det of an } (n-1 \times n-1) \\ \text{matrix with row} \\ i \text{ \& col } j \text{ removed}\end{array}\right)$$

$$\det(A) = a_{11}\det(C_{11}) + a_{12}\det(C_{12})$$

$$+ \cdots - - - - \cdots + a_{1n}\det(C_{1n})$$

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$